

Challenges related to Ducts, Chimneys and Tail-end Alignment

1. Inspection and cleaning of high-rise chimneys in paint shops

High-rise chimneys are utilized for the exhaust from painting and baking processes. These exhausts contain fumes of burnt gases, paint fumes, and volatile organic compounds (VOCs). Exhaust particles deposit on the inner walls of the chimneys, forming a flammable layer that pose a fire risk and thus require periodic cleaning. These high-rise chimneys are made of mild steel, ranging from 25 to 40 meters tall with a base diameter of 1 to 2 meters and a top diameter of 0.5 to 0.75 meters. Accessing these chimneys for inspection and internal cleaning (in case of deposition) is difficult and at times not possible.

Current methods, such as visual inspection have limitations due to inadequate camera resolution, narrow chimney diameters (limiting use of drones), and the risk of damage from exhaust fumes. Physical climbing for inspection involves high safety risks and does not allow for internal assessment. Deploying sensors that can withstand heat and gas exposure is essential, but the challenge remains in reaching the points inside the chimney to install these sensors due to height, confined space, and access issues.

We seek innovative solutions for measuring internal (material deposition and thickness) and external health (outer casing condition, rust, ladder condition, bolts condition, top canopy condition) of chimney surfaces enabling timely decision-making. Additionally, cleaning the inside of ducts as well as gas composition analysis is desirable.